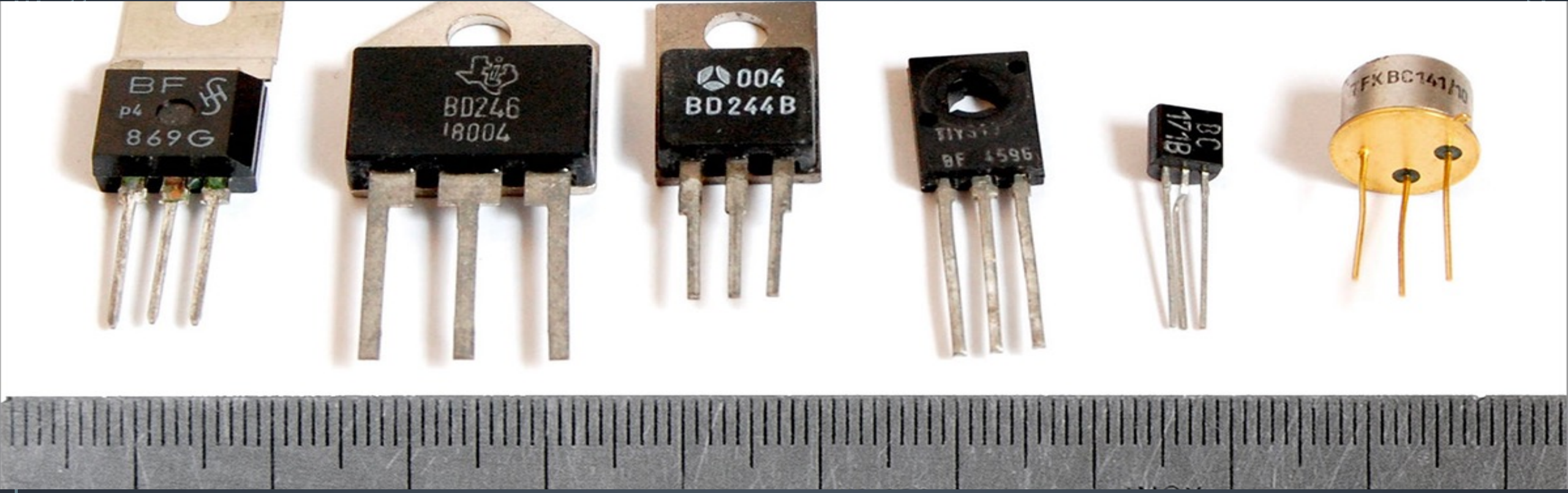


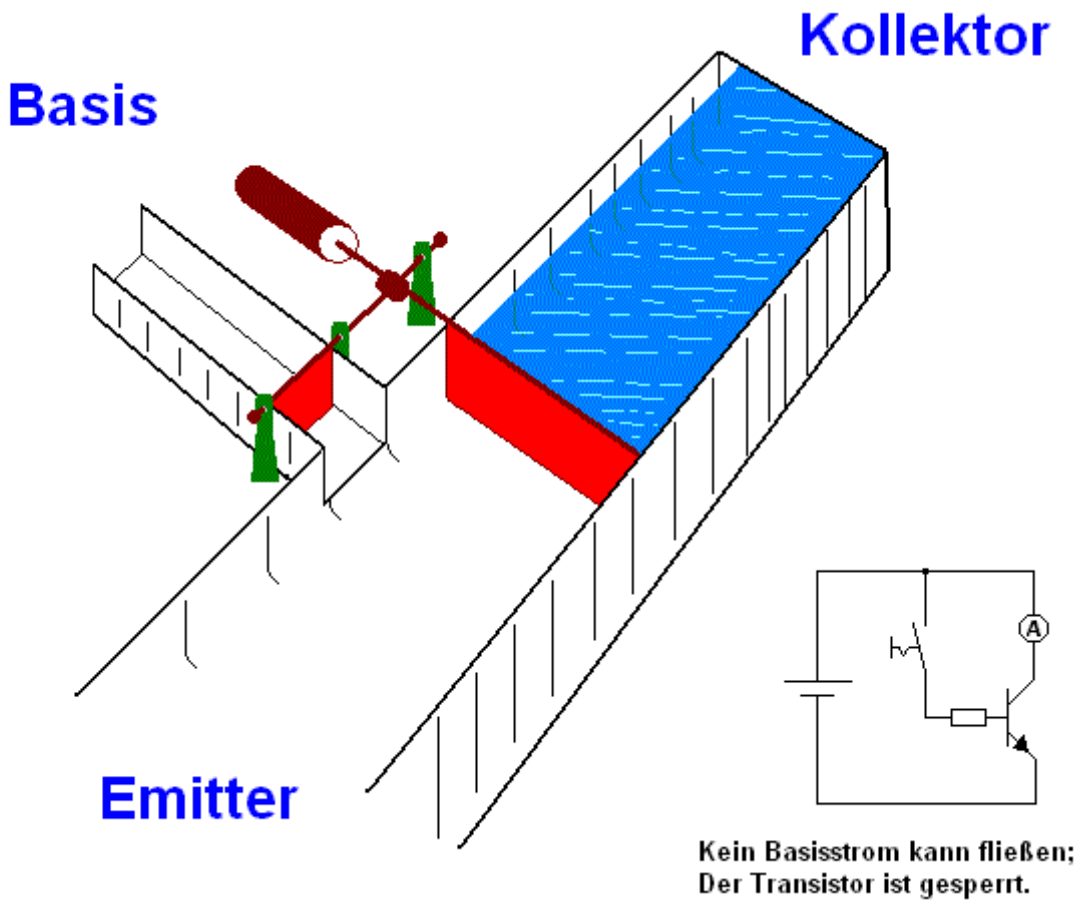
TRANSISTORS

For a non-technical audience

Fc803, jc4284, at3049, ts2859, hkc49
PW1 Group 2

CONTENTS

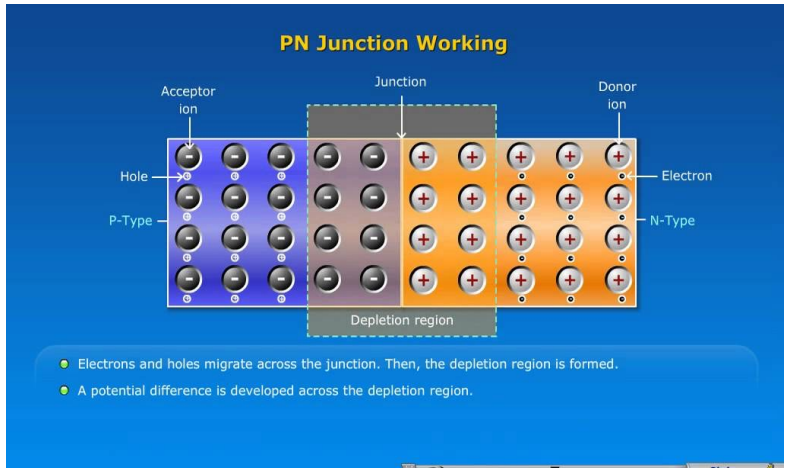
- Characteristics - why do they exist and what do they do?
- Structure – how do they work?
- Applications - what do we use them for?

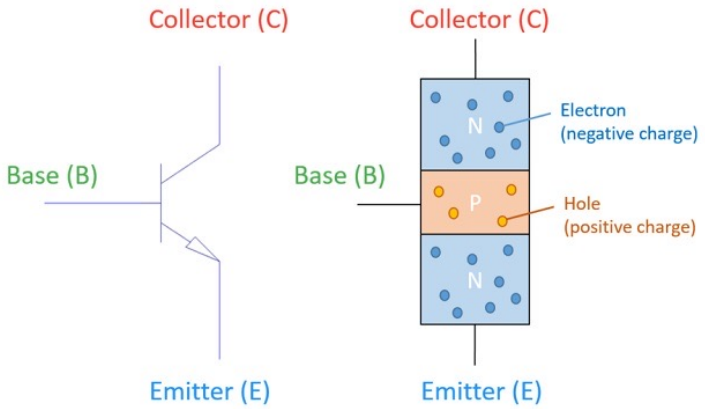


TRANSISTOR CHARACTERISTICS

- **3 terminal** device
- Allows small signals to control big ones – works as an **amplifier**
- Miniature switch – can let us easily **control signals**

This animation here shows how it can work as either

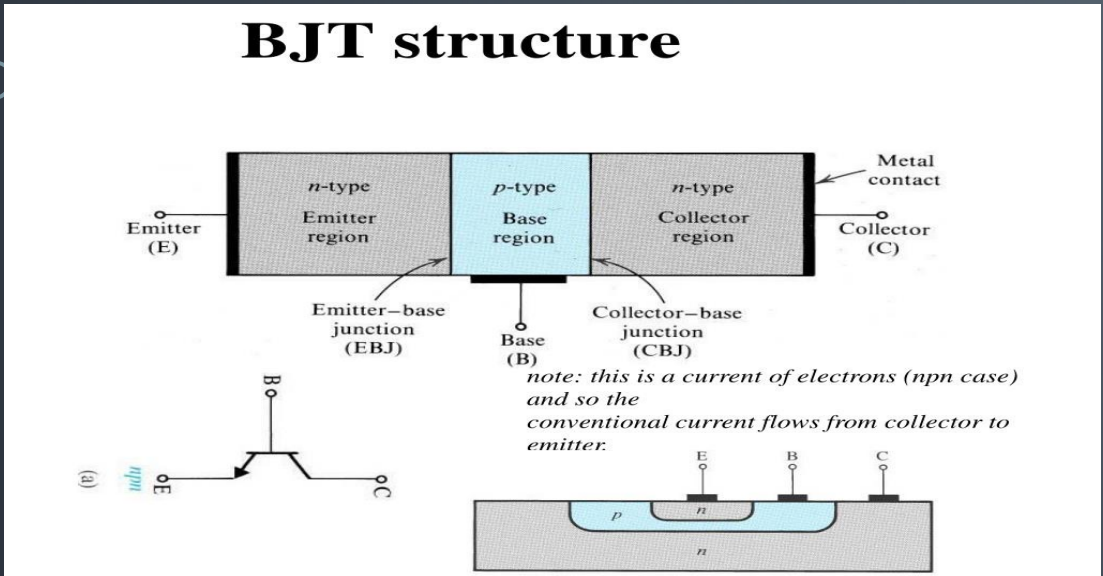


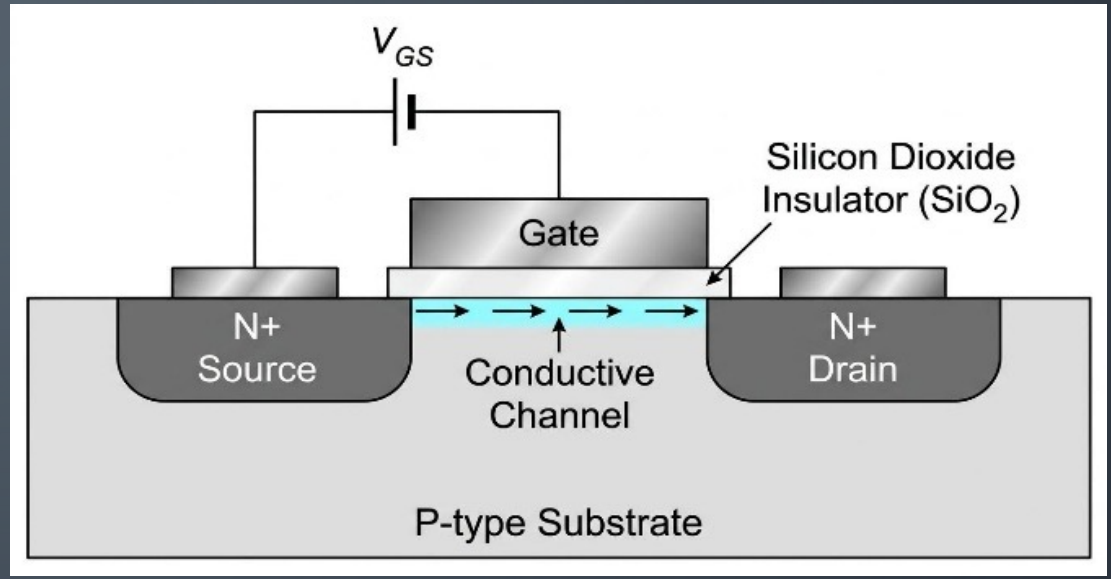


BASIC PRINCIPLES

- PN junctions are the basic building block of a transistor – they let us set up a potential difference across a semiconductor.
- By putting several of these together, we can make transistor
- Applying charge to the middle allows charge to flow through the whole thing

BJT & MOSFET





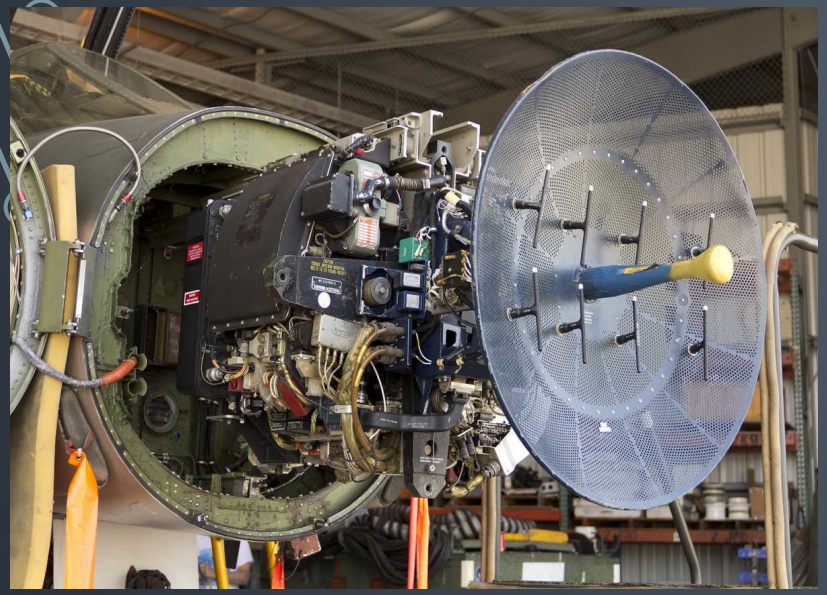
Bipolar Junction transistor

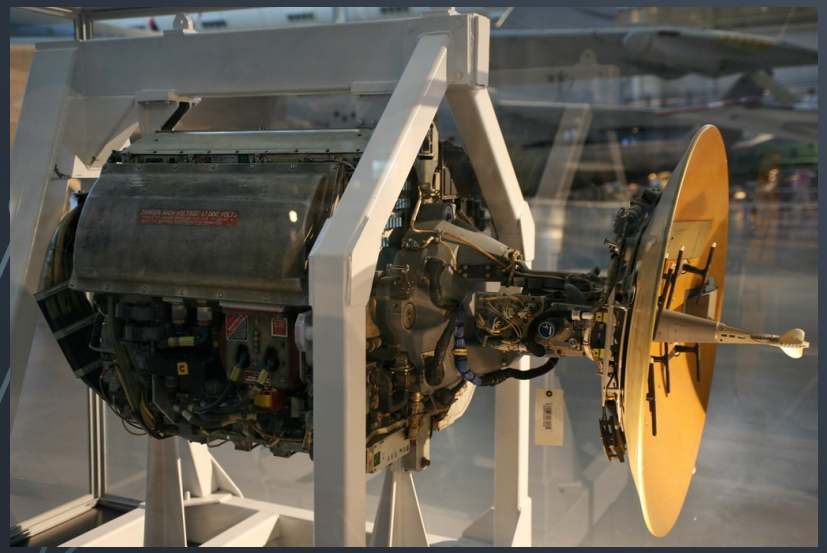
- Sandwich of junctions
- Stimulate the middle to allow charge to flow from top to bottom
- Mainly used to amplify weak signals or switch on/off high currents
- Current controlled

Metal Oxide Semiconductor Field Effect Transistor

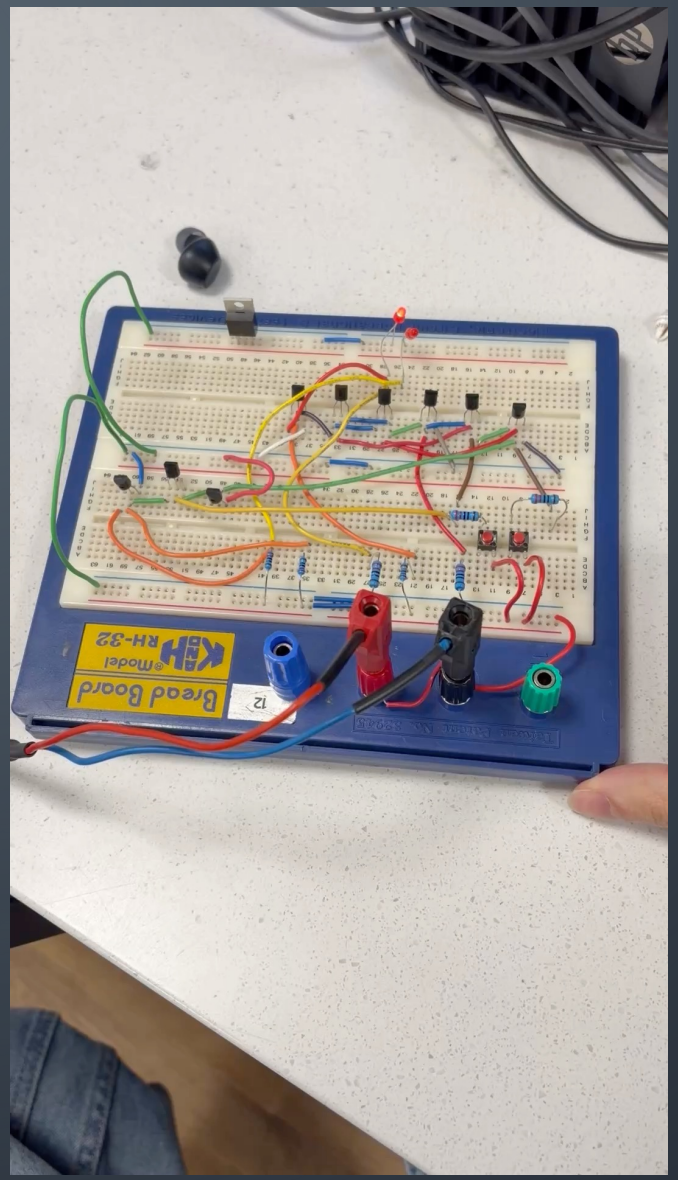
- A charged metal strip oxidises an insulator and semiconductor layer below
- Charge the strip to allow a current to flow between the two sides
- Mainly used in high frequency signals or power control
- Voltage controlled

APPLICATIONS





A valve based (top) vs transistor based radar (bottom). The transistor radar can track targets at up to twice the range for the same size



The half adder beside combines a transistor based AND and XOR gate